Independence

Suppose we are dealing with two r.v. X and Y

$$X \sim f_X, \quad Y \sim f_Y$$

We say that X and Y are **independent** if and only if

$$f_{X|Y}(x|y) = f_X(x)$$

or, equivalently[1]

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

[1] by definition $f_{X,Y}(x,y) = f_{X|Y}(x|y) f_Y(y)$

X	Y
x_1	y_1
x_2	y_2
...	...
x_n	y_n

$$\longrightarrow \quad s = s(x_1, ...x_n, y_1, ..., y_n) \quad \longrightarrow$$

$$X, Y \sim \mathcal{N}(0,1) \tag{1}$$

$$X \sim \mathcal{N}(0,1), \quad Y = a\,X + \sim \mathcal{N}(0,1) \tag{2}$$

$$\tag{3}$$

Given two sets of observations X_i and Y_i, Pearson's correlation coefficient r is defined as

$$r = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_k (x_k - \bar{x})^2} \sqrt{\sum_l (y_l - \bar{y})^2}}$$

so that $r \in [-1, 1]$.
If $X, Y \sim \mathcal{N}$ independently (null hypothesis) the pdf for r has the exact distribution

$$f(r) = \frac{\left(1 - r^2\right)^{\frac{n-1}{2}}}{\sqrt{n-2}\ \mathrm{B}\left(\frac{1}{2}, \frac{n-2}{2}\right)}$$

Given as set of observations X_i we define the **ranks** of X as

$$R(X) = \{r(x_1), r(x_2), ..., r(x_n)\}, \quad r(x_i) \in \mathbb{N}$$

so that

$$r(x_i) < r(x_j) \Leftrightarrow x_i \leq x_j \quad \text{and} \quad r(x_i) < \#X_i$$

Sperman's ρ correlation coefficient between X_i and Y_i is the **linear correlation between the ranks $R(X)$ and $R(Y)$**

$$\rho = \frac{\mathrm{Cov}(R(X), R(Y))}{\sigma_{R(X)}\sigma_{R(Y)}}$$

$$Y = \sin(X) + \mathcal{N}(\mu, \sigma^2)$$

X	Y
0.5	2.0
4.1	2.4
2.7	1.5
2.2	3.0

$\longrightarrow$

$R(X)$	$R(Y)$
1	2
4	3
3	1
2	4

$\longrightarrow$

$R(X)$	$R(Y)$	
1	2	r_1
2	4	r_2
3	1	r_3
4	3	r_4

$$\xi_n(X, Y) = 1 - \frac{3 \sum_{i=1}^{n-1} |r_{i+1} - r_i|}{n^2 - 1}$$

Theorem 1. If Y is almost surely not a constant with law μ. Then

$$\xi_n(X,Y) \quad \xrightarrow[n\to\infty]{a.s.} \quad \xi(X,Y) = \frac{\int \mathrm{Var}\left(\mathbb{E}\left[\mathbb{1}_{\{Y\geq t\}}|X\right]\right) d\mu(t)}{\int \mathrm{Var}\left(\mathbb{1}_{\{Y\geq t\}}\right) d\mu(t)} \in [0,1]$$

with $\xi(X,Y)$ being 0 if and only if X and Y are independent, and being 1 if and only if there is a measurable function $f:\mathbb{R}\to\mathbb{R}$ such that $Y=f(X)$ almost surely.

Theorem 2. If X and Y are independent and Y is continuous, then

$$\sqrt{n}\,\xi_n(X,Y) \xrightarrow[n\to\infty]{distr} \mathcal{N}\left(0,\frac{2}{5}\right)$$

Importantly, this convergence is quite fast ($n\sim 20$).

1. ξ is **non symmetric**, but can be symmetrized with

$$S_\xi(X, Y) = \max\{\xi(X, Y), \xi(Y, X)\}$$

2. **minimal assumptions**: (X, Y) is i.i.d., Y is non-constant
3. **asymptotic boundedness:** "it is clear that $\xi(X, Y) \in [0, 1]$"[2]
4. **range**: in the absence of ties

$$-\frac{1}{2} + O\left(\frac{1}{n}\right) \leq \xi_n(X, Y) \leq \frac{n-2}{n+1}$$

5. **There is a python package** with ξ and new work build on top of it

[2] One has to be careful here

$$Y = \sin(2\nu\pi X) + \cos(\nu\pi X) + \mathcal{N}(0, \sigma)$$

$$\begin{cases} X \sim \mathcal{N} \\ Y \sim \mathcal{N} \\ Z = \cos^3(X) + \mathcal{N} \\ W = Y^4 + \mathcal{N} \end{cases}$$

$$\begin{cases} X \sim \mathcal{N} \\ Y = \sin^2(X) + \mathcal{N} \\ Z = \cos^3(X) + \mathcal{N} \\ W = Y^4 + \mathcal{N} \end{cases}$$

	n	**n**
ξ_1	$X_n^{(1)}$	$Y_n^{(1)}$
ξ_2	$X_n^{(2)}$	$Y_n^{(2)}$
...	...	...
ξ_{10^6}	$X_n^{(10^6)}$	$Y_n^{(10^6)}$